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SANDEEP NADELLA

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SUMMARY

Senior Software Engineer with 11+ years of experience designing enterprise integration platforms, distributed backend systems, and cloud-native services utilizing various tech stacks involving NodeJS, Java. Experienced in event-driven architectures, APIs, performance optimization, and cross-functional technical leadership, with a growing focus on AI-powered engineering platforms.

PROFESSIONAL EXPERIENCE

Align Technology , San Jose: Senior Software Engineer **June 2020 – Oct 2026**
Consumer Platform, Global Invisalign Experience, Enterprise Engineering Teams

- Architected, Built, Owned end-to-end architecture and the delivery roadmap of fully customizable Data Integration Hub to replace TIBCO using node sandbox eliminating the need for point-to-point integrations and specialized integration engineers saving $1Million+ in operating costs.
- Built an AI-assisted playground and test-generation framework to simulate integration workflows during development, before deployment to development and staging environments.
- Developed Salesforce connectors to exchange CDC, platform events over push topics, PubSub API with Confluent Kafka topics
- Integrated Apache Avro schema support into Data Integration Hub connections.
- Proposed failed-event reprocessing using sandbox exceptions propagated to stage-level handlers and asynchronous database logging of failures.
- Automated integration-workflow code generation with AWS Bedrock models; built a Model Context Protocol (MCP) server enabling local LLMs to construct no-code integrations from user inputs.
- Built Data Integrity Check dashboards for identifying and conducting large scale data corrections on production systems through AI assisted coding using Cursor Composer, Claude Code and integrated with Large Language Models(LLMs) in AWS Bedrock for mappers.
- Designed and built fault-tolerant order management service for processing, fulfilling orders placed in CPQ systems like Bit2Win. Designed Model Context Protocol(MCP) server for accessing read-only metrics.
- Designed and built GraphQL schemas, resolvers, used dataloaders, RESTful and Non-RESTful APIs, Kafka publish-subscriber models for various features spanning across Virtual Care, Subscriptions Portal, Align Special Markets Portal, Align Diagnostics Subscription, Align Admin Back Office, My Invisalign Web, iOS, Android applications
- Designed Subscription-based payment models for use in Vivera Retainers, Align Diagnostics, and Treatment Planning at Invisalign offered by 100K+ doctors while working with global teams
- Stabilized, and maintained GraphQL, NodeJS, NestJS using Prisma ORM and Ruby on Rails code repos. Identified architectural risks early in the development lifecycle, proposing design improvements to reduce technical debt and future rework.
- Significantly contributed towards Ruby on Rails, NodeJS microservices at the API, Schema, and System Design levels. Modernized tech stack from RoR to Node.
- Improved performance of API endpoints by over 40% in some cases across services through SQL, Application layer(ActiveRecord), and API design optimizations, and through requirement cut-downs when necessary while ensuring system scalability and built backends to serve metric reports
- Supported deployment of docker container stack over Heroku, AWS through Bamboo CICD ensuring Checkmarx, Snyk Security fixes, Unit Test Case integrations. Reported and fixed several security issues in services across Align.
- Guided and mentored Developers, QA, Analysts, PMs across experience levels over project releases and ensured product growth, successful timely deliveries. Proposed scalable, fault-tolerant, maintainable designs across services.
- Built point-to-point integrations involving Google Ad forms and Meta Lead Ad forms(Facebook and Instagram) with Align Lead Service API
- Designed GraphQL resolvers, schemas, backend APIs and assisted in the development of the Aligner fit feature
- Responsible for incorporating GraphQL integration test framework to ensure high-quality deliverables avoiding regressions
- Built monitoring dashboard application for analyzing the performance metrics of applications deployed on Heroku. Tech stack includes React, NodeJS, Postgres and Redis.
- Led Align team members and contractors in technical discussions, took ownership of investigations, and developed solutions to identify and fix production issues.

Arizona State University, Tempe: Full Stack Software Engineer **February 2019 – May 2020**

- Designed and built Analytics applications for Student grades for ASU SoMSS, AMLSS departments, MTBI program using Node.js, Angular, Cube.js and deployed in AWS
- Developed core application features including JSON Web Token(JWT) based application login, Secure file storage in AWS S3, Visual analytics dashboard using Cube.js and Chart.js

- Deployed applications on Amazon Web Services(AWS) and Apache web servers
- Deployed and Maintained Docker-based Open source software such as Opencast for SoMSS, Concerto for CHS departments
- Integrated Concerto with REDCap(Clinical EDC Software for translational research) for the College of Health Solutions department using R language
- Offered technical assistance for the maintenance of High-performance Computing servers, Department and Lab servers involving Bash scripting

Oracle Corporation, India: Senior Software Developer **June 2015 – July 2018**
Commerce Cloud Integrations Team

- Led the following integration projects
  - Commerce-Responsys: Orders, Profiles and Abandoned Cart sync flows
  - Commerce-SiebelCRM: Order sync flows
  - Commerce-Oracle Engagement Cloud for Communications: Order API and Webhook flows
  - Commerce-ProductHub: File Upload/Download, Products Import/Export flows
- Developed Commerce Cloud Adapter by working on Oracle Integration Cloud Services (ICS) SDK
- Implemented and maintained Swagger functionality of ICS/OIC CC adapter
- Designed XSLT templates for mapping data for Oracle FOM, OROMS and Commerce and implemented Product Hub integration
- Formulated Swagger template based NodeJS custom extensions development for Commerce Cloud
- Led the successful migration of Commerce Cloud API documentation to JSON Hyper-Schema, OpenAPI Specification
- Contributed to Commerce-Responsys, Commerce-SiebelCRM, Oracle Engagement Cloud for Communications integrations
- Developed the front end and back end designs for CC-OROMS integration using Java and JavaScript

EDUCATION

Master of Computer Science (MCS), Computer Science **August 2018 – May 2020**
Arizona State University (ASU), Tempe; United States GPA: 4/4

Bachelor of Technology (B.Tech) **July 2011 – May 2015**
National Institute of Technology, Warangal (NITW); India GPA: 8.12/10 (≈ 3.6/4)

ACHIEVEMENTS

- Recognized by business, management and customers for exceptional technical leadership in architecting scalable enterprise systems, influencing critical design decisions, taking ownership of production issues, coordinating with internal teams and mentoring colleagues and contractors, and keeping stakeholders informed through resolution.
- Earned recognition for adapting quickly to changing priorities while collaborating across teams to address business and customer needs. Received several Align Achiever awards for Accountability, Collaboration and Agility.
- Engineering Graduate Fellowship **2018-2019, 2019-2020**

TECHNICAL SKILLS

Technical Leadership, System Design, Algorithms, Analytical, Critical thinking skills, Mentoring
Programming Languages: Java, JavaScript, SQL, Python
Technologies, Libraries, Tools: Node.js, React.js, Prisma ORM, Kubernetes (K8s), Claude Code, HTML5, CSS, D3.js, Cube.js, Chart.js, Bootstrap, Angular (TypeScript), JSON Web Tokens (JWT), OAuth, Azure, Spring Boot, Oracle ATG Web Commerce platform, J2EE and Web design, development essentials, REST and SOAP web services, JSON/XML, Claudia.js, Amazon Web Services (AWS), AWS Elastic Kubernetes Service(Amazon EKS), AWS Lambda, AWS Bedrock, API Gateway, Bamboo CI/CD, OpenSearch Observability, Android App Development SDK, MATLAB, XSLT, Jasmine testing, Selenium IDE, JUnit testing, WebRTC, MySQL, PostgreSQL, MariaDB, MongoDB/NoSQL, Python – Django framework, TensorFlow APIs, Git, Perforce code version control tools, Gradle, Maven build tools, OpenCV, Docker, Google Firebase, ML Kit, Spotify SDKs, Figma/Axure RP for Prototyping